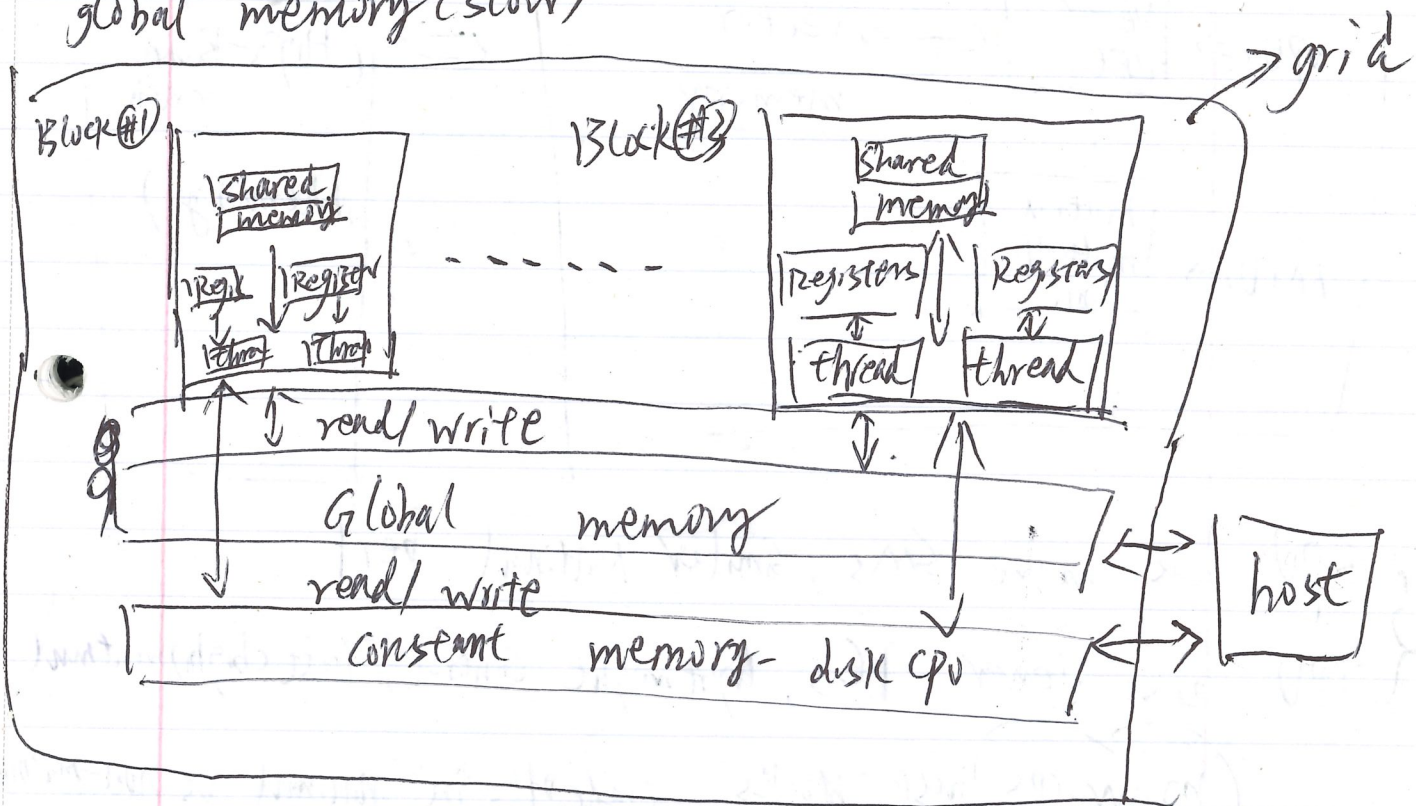


Memory model of a GPU

Each thread can access its own register and shared memory within the block

Information that goes across blocks need to read/write to global memory (slow)



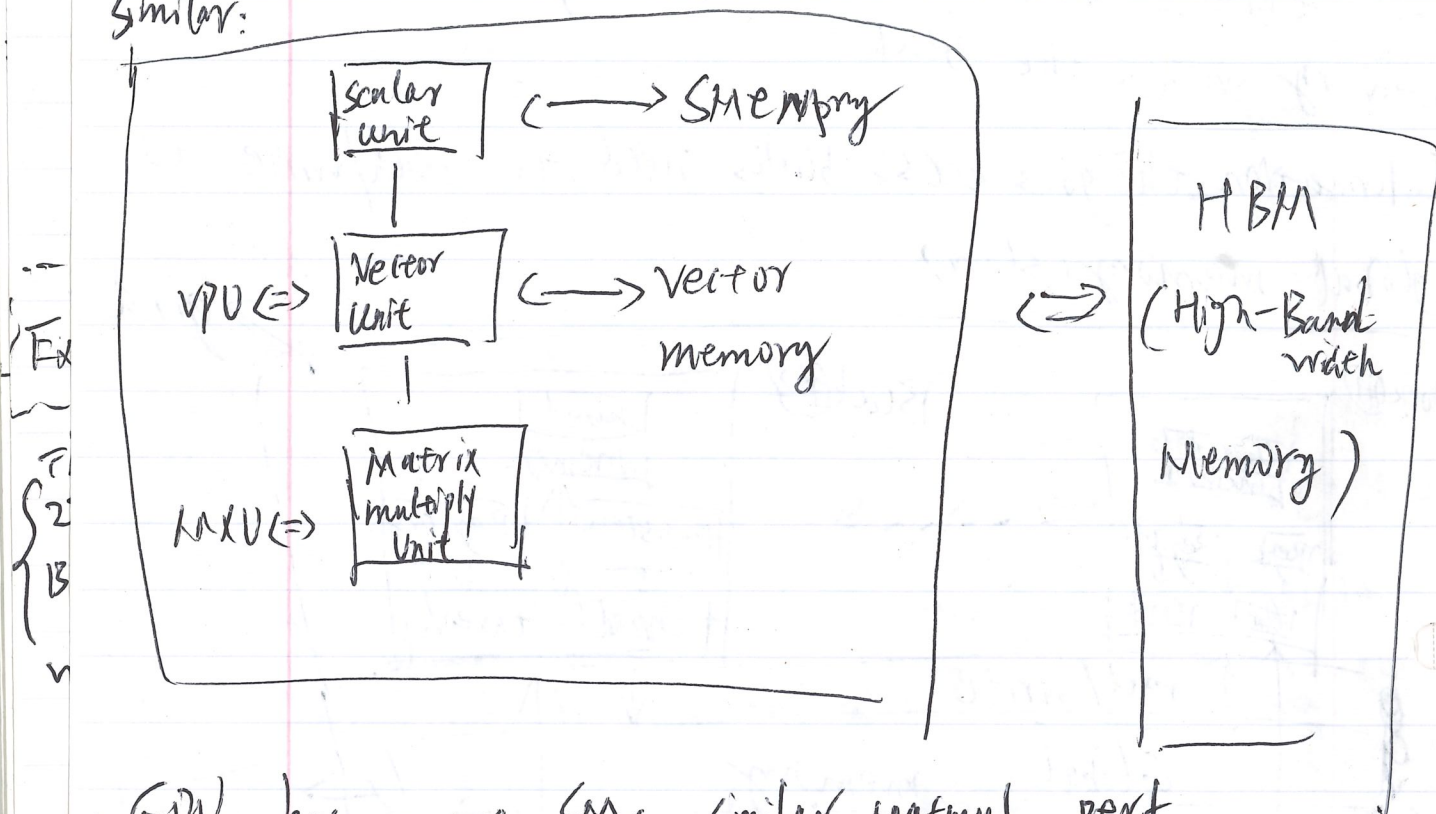
Device code can:

- read/write - R/W per-thread registers
- R/W per-thread local memory
- R/W per-block shared memory
- R/W per-grid global memory
- Read only per-grid constant memory

Host code ~~can~~ can do: Transfer data to/from per grid global and ^{constant} memory

what about TPU? (not GPU)

GPUs, TPUs and many other accelerators are at high-levels similar:



GPU has more SMs, similar matmul perf

TPU has fewer TCs, lightweight control, fast (high) matmul,

no warps, just blocks - trade offs in matmul vs non-matmul

